Project 4: Cross-validated estimation in surveys

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1 Introduction

For this project, we examined the influence of different models on various types of mean estimation through cross-validation. Specifically, we tested linear regression, regression trees, and splines for model-assisted estimation and imputation. For reweighting we use logistic regression, decision trees and random forest. Where model-assisted estimation was used on a complete data set, for the testing of imputation and reweighting we implemented a data set with missing points. The goal of this project was to select the estimator that minimizes the cross-validation error.

Question (as in the project document):

- How does the selected mean estimator, which is selected as the estimator that minimizes the overall cross-validation error, perform compared to the other estimators?
- This question is to be answered for model-assisted estimators, imputation estimators, and reweighted estimators

2 Methods

2.1 Data

To examine the question, we created a data set in R for calculating the average grade of 1000 university students using a sample of 250 students. Our covariates are year of study, number of hours spent studying per week, number of classes, and number of nights students go out per week. Here is a breakdown of how each variable in the dataset was created:

- Year of study: 1000 random integers between 1 and 5 derived from a gamma distribution with shape = 2 and scale = 1.2. For this covariate, we placed a cap and a floor, as well as rounded up to the nearest integer.

- Weekly hours of study: 1000 random numbers rounded to 2 decimal places between 0 and 49 drawn from a normal distribution centered at 25 with a standard deviation of 10.
- Number of classes: 1000 random integers between 1 and 7 drawn from a normal distribution centered at 4 with a standard deviation of 1.5 (rounded to the nearest integer).
- Weekly nights out: 1000 random integers between 0 and 7 derived from a normal distribution centered at 3.5 with a standard deviation of 1.5. The values from the distribution were rounded to the nearest integer for the final data set.
- Grade (variable of interest): a linear function of the covariates rounded to 2 decimal points, floored at 0, and capped at 100. The function is $25 + 2 * \text{year.of.study} + 1.5 * \text{study.hours} + 3 * \text{number.of.classes} - 2 * \text{nights.out} + \epsilon$ where ϵ is a vector of noise (1000 random numbers from a normal distribution with mean 0 and standard deviation 5)

From our dataset, the true mean grade is calculated to be 72.51394.

2.2 Model-Assisted Estimation

The mean estimator for model-assisted estimation is as follows:

$$\hat{\mu}_{ma}(f) = \frac{1}{N} \sum_{k \in U} \left\{ f(\mathrm{x}_k) + \frac{I_k}{\pi_k} \times (y_k - f(\mathrm{x}_k)) \right\}$$

2.2.1 Linear Regression

To observe model-assisted estimation with a linear regression model, we first randomly sampled 250 students from the full data set. Then, we fit a linear model to predict the grades of the students based on covariates, and used this information to estimate the population mean.

2.2.2 Regression Trees

Following a similar procedure, for the regression trees, instead of implementing a linear model, we fit a regression tree on the sampled data.

2.2.3 Splines

Again, following a similar procedure, we used splines to model the relationships between the covariates and the grades.

2.2.4 Cross-Validation and Methods for Analysis

For cross-validation, we used Monte-Carlo simulations (R=1000) to evaluate and compare the three models. Within each iteration, we created empty vectors to

store the cross-validation (CV) error for five fold of each model. Four folds were used as training with 1 fold for testing. The cross-validation estimator was selected based on the lowest cross-validation error in each iteration. We tracked which model performed best in each iteration and also computed the bias and mean squared error (MSE) for each model over all simulations.

2.3 Imputation

The mean estimator for imputation is as follows:

$$\hat{\mu}_{imp}(f) = \frac{1}{N}\left(\sum_{k\in S_r}\frac{y_k}{\pi_k} + \sum_{k\in S_m}\frac{f(\mathrm{x}_k)}{\pi_k}\right)$$

For imputation, we created a vector of non-respondents using Bernoulli sampling and randomly selecting 250 data points. We imitate missing at random data by masking the grades for half of the selected 250 students (using the Bernoulli sample to indicate which are masked).

The procedure we used for imputation with linear regression, regression trees, and splines follows the procedure used for model-assisted estimation, but using the data with non-respondents. Because of this, the models are trained only on the respondents' data and then used to impute grades for the nonrespondents. Thus, for the cross-validation we only run the CV on respondents. Otherwise, like with the model-assisted estimation, we use the same models, and apply the CV error across 5 folds over 1000 Monte-Carlo simulations, computing the average CV error, bias, and MSE for each method.

2.4 Reweighting

The mean estimator for reweighting is as follows:

$$\hat{\mu}_{ipw}(p) = \frac{1}{N}\sum_{k\in S_r}\frac{y_k}{\pi_k p(\mathrm{x}_k)}$$

Since we are not dealing with continuous data with reweighting (1 for respondents and 0 for nonrespondents), we decided that linear regression, regression trees, and splines were not the right models, and instead used logistic regression, decision trees, and random forest. We simulated nonresponse using a binary response indicator for each unit (using Bernoulli, as with imputation), and then identified the respondents within a randomly selected sample. Using inverse probability weighting, we computed the population mean of the outcome. Similar to the past two estimation methods, we used Monte-Carlo simulations (1000) and 5-fold CV to assess the classification accuracy of each method for predicting response status, calculating the average MSE, average Bias, and average CV accuracy.

3 Results

3.1 Model-Assisted Estimation

Model	Model Type	$\hat{\mu}_{ma}(f)$	Average MSE	Average CV Error
$\hat{m}_1$	Linear Regression	73.03458	0.2710648	26.27172
$\hat{m}_2$	Regression Trees	72.60471	0.008238684	65.03227
$\hat{m}_3$	Splines	73.11977	0.3670259	27.3289

Table 1: Summary of results for model-assisted estimation, including a sample $\hat{\mu}_{ma}(f)$ value (changes slightly with each iteration), as well as the averaged MSE and CV Error of the estimator for each model.

Linear Regression	Regression Trees	Splines
926	0	74

Table 2: Number of times each model was selected as the best model over the 1000 simulations for model-assisted estimation.

3.2 Imputation

Model	Model Type	$\hat{\mu}_{imp}(f)$	Average MSE	Average CV Error
$\hat{m}_1$	Linear Regression	72.61132	0.009483177	30.2303
$\hat{m}_2$	Regression Trees	72.61132	0.0003816805	84.86446
$\hat{m}_3$	Splines	72.55503	0.001688784	33.84927

Table 3: Summary of results for imputation, including a sample $\hat{\mu}_{imp}(f)$ value (changes slightly with each iteration), as well as the averaged MSE and CV Error of the estimator for each model.

Linear Regression	Regression Trees	Splines
972	0	28

Table 4: Number of times each model was selected as the best model over the 1000 simulations for imputation.

3.3 Reweighting

Model	Model Type	$\hat{\mu}_{ipw}(f)$	Average MSE	Average CV Accuracy
$\hat{m}_1$	Logistic Regression	72.96933	0.8294773	0.488604
$\hat{m}_2$	Decision Trees	73.32955	3.731753	0.496428
$\hat{m}_3$	Random Forest	56.16915	304.0661	0.000196

Table 5: Summary of results for reweighting, including a sample $\hat{\mu}_{ipw}(f)$ value (changes slightly with each iteration), as well as the averaged MSE and CV Error of the estimator for each model.

Logistic Regression	Decision Trees	Random Forest
450	550	0

Table 6: Number of times each model was selected as the best model over the 1000 simulations for reweighting.

4 Analysis

4.1 Model-Assisted Estimation

As is evident in Table 1, though completing model-assisted estimation with regression trees has the lowest MSE and Bias, it has the highest CV error by far. Thus, as can be seen in Table 2, it is never selected as the best model over the 1000 simulations. Though the linear regression model has a higher bias and MSE, it has the lowest average CV error and is selected as the best model 92.6% of the time. Thus, linear regression is the best model for model-assisted estimation with our data set.

4.2 Imputation

In comparison to model-assisted estimation, imputation consistently produces lower bias and MSE, which is interesting considering the missing data points in imputation that are present for model-assisted estimation. Similar to with model-assisted estimation, in the case of imputation, regression trees, though producing the lowest bias and MSE for the estimator, are never selected as the best model based on CV error. Additionally, linear regression is proven to be the best model for imputation, as can be seen in Table 4 and by the CV error in Table 3.

4.3 Reweighting

For the case of reweighting and the different models, logistic regression and decision trees are both adequate models to use based on the CV error. Evidently, random forest is not a good modeling technique for reweighting based on our

simulations as it produces the largest average bias and average MSE. In contrast to model-assisted estimation and imputation, we looked at the average CV accuracy rather than the average CV error. Thus, we look for the largest value rather than the smallest. Where logistic regression and decision trees produce similar CV accuracy, random forest has a very small accuracy rate. Thus, it is not surprising that this model is never selected as the best model based on cross-validation in our simulation. Notice that the accuracy rate for decision trees and logistic regression is relatively low. This is because when we created the non respondents, we made them follow a Bernoulli distribution with probability of success 0.5. Thus, the probability of selecting a respondent is equal to the chance of selecting a non respondent. Therefore, when fitting a probability distribution function with covariates, it is evident that a good model will get an accuracy rate close to 0.5.

5 Conclusion

In the case of model-assisted estimation, the selected mean estimator is the linear regression model, and it well outperforms the other models we simulated in terms of cross-validation error.

Similarly, for imputation, the mean estimator used for a linear regression model is the best compared to regression trees and spline spectra when observing the cross-validation error. This model is selected most of the time in simulations.

Finally, for reweighting, the mean estimator that performs best based on our simulations is the one used with decision trees. However, this model does not significantly outperform logistic regression, so using logistic regression for estimation with reweighting is reasonable based on the overall cross-validation error.